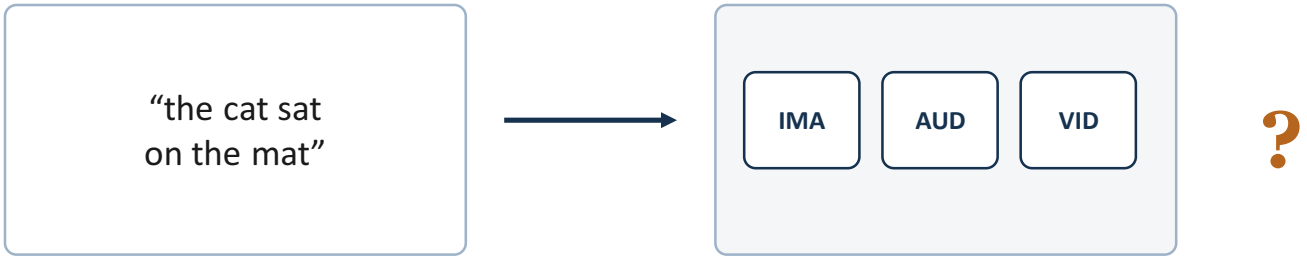
MULTIMEDIA IR

Data Modeling

A Different Kind of Document



Boolean, Vector, Probabilistic — all of it assumes the document is made of words. This one isn't.

Three Dimensions of a Multimedia Object

Form

the raw file itself

Content

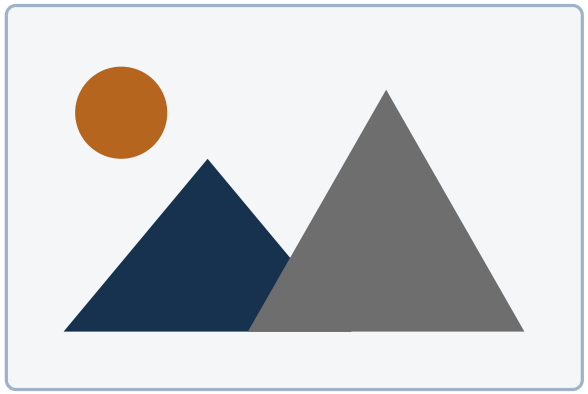
what it depicts or means

Structure

how it's built from parts

Every real multimedia system has to answer all three before it can be searched.

Form



`.jpg` `.png` `.raw`

same photograph, three different raw representations

The physical, digital encoding — format, resolution, bitrate. Two systems can't even exchange a file without agreeing on this.

Content — Two Very Different Levels



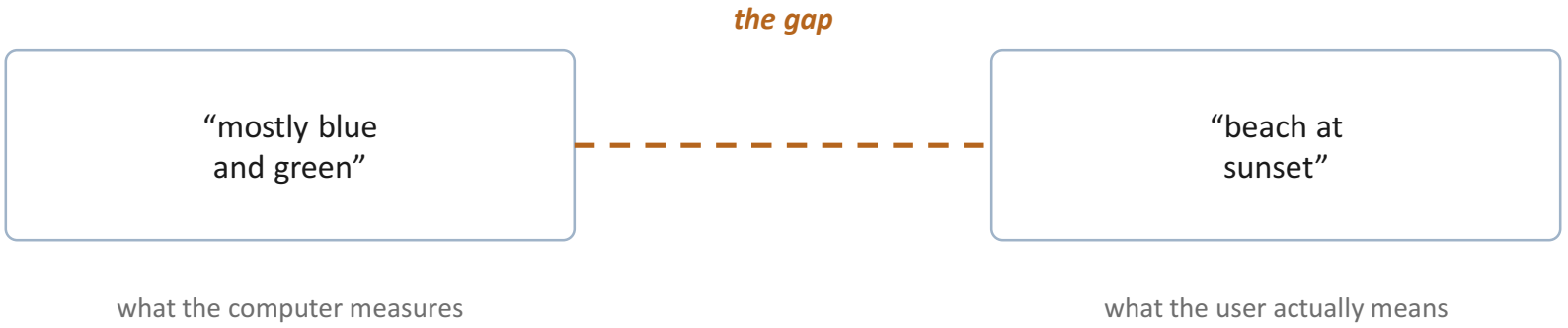
low-level: colour histogram, texture, shape

“a beach at sunset”

high-level: what a human would say it's about

A computer can compute the first directly. It cannot compute the second — that has to be inferred, tagged, or learned.

The Semantic Gap



Almost every real multimedia system exists specifically to narrow this gap.

How It's Actually Modeled



Matching becomes similarity search — closest vectors — not exact keyword match.

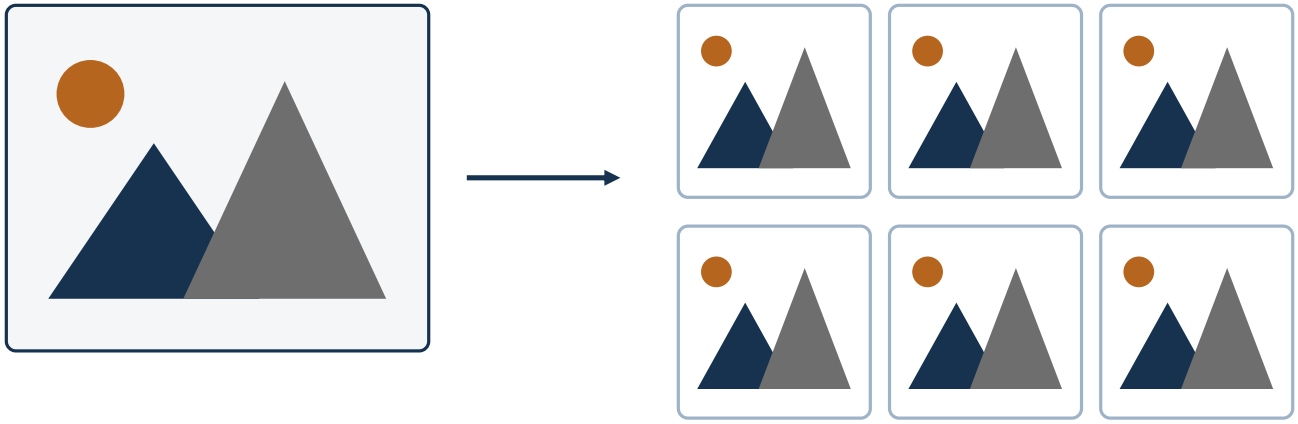
Real Life: Shazam



It never compares raw audio — it extracts a fingerprint and matches that against millions of stored fingerprints.

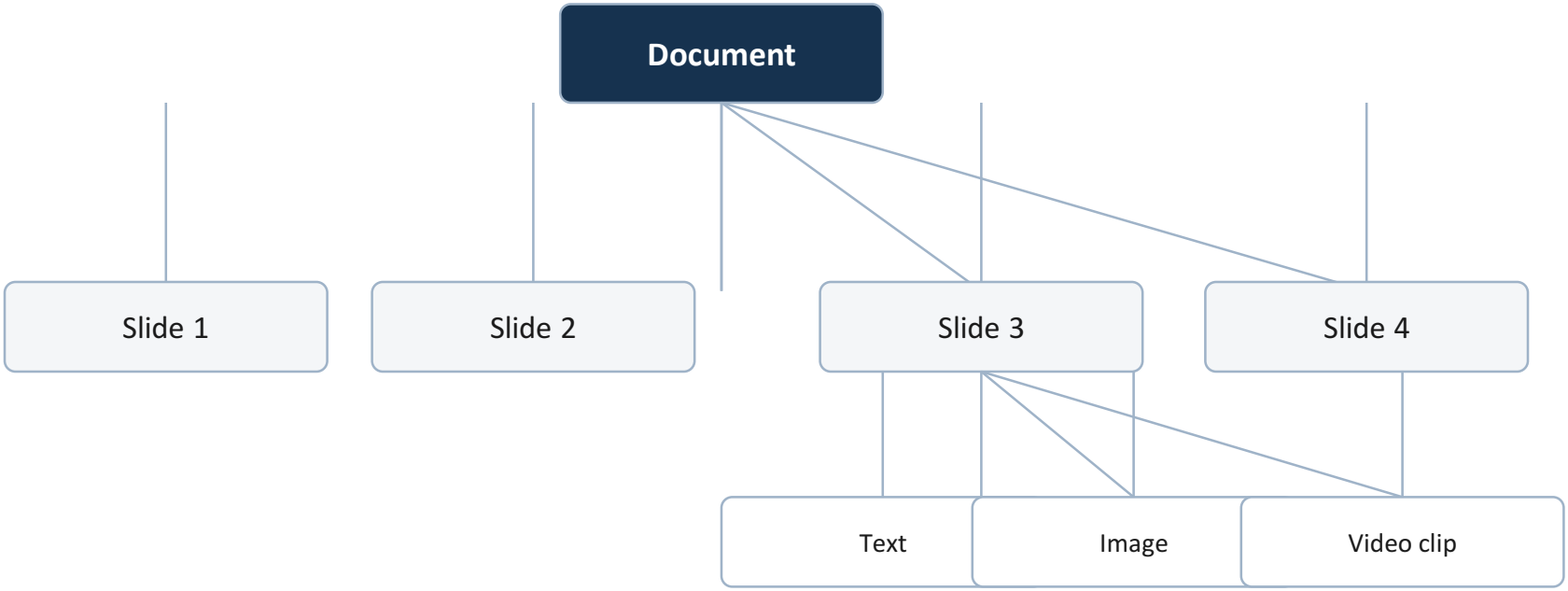
GOOGLE LENS

Real Life: Reverse Image Search



No text query at all — the image itself is converted to a feature vector and matched against millions of others.

Structure — Multimedia as a Composed Object



The same idea you already met in Structured Text Retrieval — now the parts can be any media type, not just text.

Logical vs. Layout Structure



One document, one logical structure — but it can be laid out differently on screen, in print, or on mobile.

Real Life: Hospital Imaging (PACS / DICOM)



Patient ID: _____

Modality: MRI

Body part: Knee

Date: _____

Every scan is an image object bundled with structured metadata — a real, everyday extension of a plain relational database.